

CODEX-ONLY BUILD RECORD

PlotTea**Files** PlotPipe Build Summary

A focused summary of what Codex built and tested in the PlotTeaFiles pipeline before the later Claude-updated working tree changes.

June 6, 2026

COMMITTED BUILD WORK

3 commits

TRACKED CODEX MILESTONES

Pipeline + Studio

CLI, REMOTION, NEXT.JS

01 - SCOPE

What This PDF Covers

This document only covers the updates Codex made while testing and building the PlotTeaFiles / PlotPipe pipeline. It is based on the committed repository history and the project memory that was created during that work.

Current uncommitted edits in `src/app/api/jobs/store.ts` , `src/app/page.tsx` , `src/app/styles.css` , `src/cli.ts` , `src/pipeline/compose.ts` , `src/pipeline/imagegen.ts` , and `src/remotion/components/Captions.tsx` are not counted here because you said those files and folders were updated by Claude.

Codex's committed milestones

COMMIT	TITLE	WHAT CODEX ADDED
207be4c	Initial PlotTeaFiles video pipeline	Built the first TypeScript CLI, parsing, TTS, captions, image/B-roll handling, Remotion composition, render command, sample story, docs, and project memory.
1399628	Studio web app MVP	Added the Next.js Studio UI, job APIs, in-memory job store, local/mock/remote runner modes, download endpoint, and hosted-studio planning docs.
766e629	Plain script mode to Studio	Added a script-conversion API and Studio controls so a plain English story could be converted into structured JSON before rendering.

Generated PDFs and later untracked documentation files are separate from the Codex build milestones unless explicitly listed as output artifacts.

02 - CORE PIPELINE

The Pipeline Codex Built

Codex built PlotPipe as a local Node.js / TypeScript pipeline that turns a PlotTeaFiles story script into a vertical MP4. The pipeline validates the story, creates narration, generates word-level captions, prepares visual scenes, and renders with Remotion.

Script parsing

Added Zod validation for JSON and TypeScript story scripts.

Voiceover

Integrated OpenAI TTS by default, with ElevenLabs kept as an optional provider.

Captions

Used OpenAI Whisper to create word-level timestamp data for karaoke-style captions.

Image/B-roll system

Started with local assets and Pexels, then moved toward generated AI images for better story match.

Remotion render

Created React-based video composition, captions, hook screen, vignette, B-roll motion, and MP4 rendering.

CLI commands

Added commands for full runs plus separate voiceover, subtitles, compose, and render stages.

Main files Codex created or updated

- `src/cli.ts` - CLI entry point, commands, options, and pipeline orchestration.
- `src/pipeline/parse.ts` - story schema and validation.
- `src/pipeline/voiceover.ts` - OpenAI / ElevenLabs narration generation.
- `src/pipeline/subtitles.ts` - Whisper transcription and word timestamps.
- `src/pipeline/compose.ts` - converts story, audio, captions, and visuals into Remotion props.
- `src/pipeline/render.ts` - Remotion render handling.
- `src/pipeline/imagegen.ts` - AI image generation fallback path for stronger visuals.
- `src/remotion/*` - the video composition and visual components.

03 - STUDIO

The Web Studio Codex Added

After the CLI pipeline was working, Codex added a Next.js Studio so the pipeline could be used through a browser instead of only PowerShell commands.

Browser editor

Added a UI for pasting and editing story JSON.

Job submission

Added API routes under `src/app/api/jobs` for creating and reading render jobs.

Progress tracking

Tracked queued, running, done, and failed states in an in-memory job store.

Download endpoint

Added a route for downloading rendered MP4s when a job finishes.

Runner modes

Supported `mock`, `local`, and `remote` Studio execution modes.

Plain script mode

Added conversion from plain story text into structured StoryScript JSON.

Studio files Codex touched

- `src/app/page.tsx` and `src/app/styles.css` - Studio UI and styling.
- `src/app/api/jobs/*` - job routes, job schema, job store, status route, and download route.
- `src/app/api/scripts/convert/route.ts` - plain script to structured JSON conversion.
- `src/app/layout.tsx`, `next.config.mjs`, `next-env.d.ts` - Next.js app setup.
- `docs/PHASE_3_HOSTED_STUDIO.md` - hosted Studio plan and production direction.

04 - TESTING WORK

What Codex Tested While Building

The testing work was practical: build the pipeline, run it against the sample story, inspect generated outputs, compare visual strategies, and adjust the workflow based on the results.

TEST / OUTPUT	WHAT IT PROVED	RESULT
out/tornado.mp4	Confirmed the basic Remotion render path could produce an MP4.	Worked, but visuals were placeholder-like.
out/tornado-pexels.mp4	Tested stock-video B-roll from Pexels.	Worked technically, but visuals felt generic.
out/tornado-pexels-v2.mp4	Tested improved Pexels keyword refinement.	Better search behavior, still not strong enough.
out/tornado-pexels-photos.mp4	Tested photo-only stock output as an alternative.	Still too stock-like for the channel style.
out/tornado-ai.mp4	Tested AI-generated image visuals.	Best direction compared with stock footage.
out/tornado-gemini.mp4	Tested Gemini as the image provider.	Gemini hit 429 Too Many Requests ; OpenAI fallback generated the usable images.
out/tornado-ai-cinematic.mp4	Tested multi-shot AI images per segment with cinematic motion.	Best result from the Codex testing phase.

The testing conclusion was that Pexels did not match the dramatic story format well enough. The strongest approach was AI-generated cinematic stills, multiple shots per segment, slow motion, cuts, grain, vignette, and captions over narration.

05 - FILES AND FOLDERS

Codex Build Footprint

These are the main areas Codex added or changed during the committed build sessions.

Source folders

- `src/pipeline/` - parse, voiceover, subtitles, compose, render, Pexels, image generation, paths.
- `src/remotion/` - Remotion entry, story composition, B-roll, captions, hook, vignette, and shared types.
- `src/app/` - Next.js Studio UI and local API routes.
- `src/utils/` - audio helpers, logging, and retry helpers.
- `scripts/` - sample story data.
- `assets/broll/` - local B-roll placeholder folder.

Project and documentation files

- `package.json` and `package-lock.json` - dependencies and scripts for CLI, Next.js, and Remotion.
- `tsconfig.json` and `tsconfig.cli.json` - TypeScript configuration for app and CLI builds.
- `remotion.config.ts` - Remotion configuration.
- `.env.example` - safe environment-variable template.
- `.gitignore` - ignores secrets, generated outputs, caches, and dependencies.
- `README.md` - setup, commands, pipeline notes, and Studio usage.
- `PROJECT_MEMORY.md` - running project memory from the build and tests.
- `docs/PHASE_3_HOSTED_STUDIO.md` - plan for a hosted Studio architecture.

Generated output folders used during testing

- `tmp/` - cached audio, subtitle, generated image, prompt, and Remotion prop files.
- `out/` - rendered MP4 test outputs.
- `public/plotpipe/` - public assets copied for Remotion rendering.

06 - FINAL SUMMARY

Exactly What Codex Did

Codex built the first complete PlotTeaFiles video pipeline, tested it through multiple visual approaches, and then wrapped it in a browser-based Studio MVP.

- Created a working CLI called `plotpipe`.
- Connected story parsing, OpenAI TTS, Whisper captions, B-roll/image handling, and Remotion rendering.
- Produced and compared several MP4 test renders.
- Learned that stock footage was not good enough for the channel style.
- Moved the project toward AI-generated cinematic stills as the best visual strategy.
- Added a Next.js Studio with job APIs, progress states, downloads, and runner modes.
- Added plain-script conversion so stories could start as normal text instead of only JSON.
- Documented setup, usage, safety notes, project memory, and hosted-Studio direction.

Generated on June 8, 2026. Scope: Codex-authored committed build work only; Claude's later uncommitted working tree changes are intentionally excluded.